

UNIVERSITY OF NEW SOUTH WALES



《SignalYield》

Systematic Multi-Asset Funds with News-Sentiment Analytics

FINS5545 Individual Project B

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Executive summary

SignalYield is an educational investment product prototype for users with basic financial knowledge. It presents the performance of systematic equity, cryptocurrency and multi-asset funds in a clear and standardised form. Based on the verified Part A data, the product constructs 12 simulated investment funds, a sector news-sentiment index and an interactive dashboard. All funds use a monthly rebalanced, expanding-window backtest: information available at each monthly decision date determines target weights that become effective on the next trading day; holdings then drift with realised returns until the next rebalance.

In terms of risk-adjusted performance, equally weighted equity funds outperformed: an annualized Sharpe ratio of 0.86, an annualized return of 13.3%, and a maximum drawdown of -20.2%. While cryptocurrency funds showed impressive returns, their volatility and drawdowns were equally significant; therefore, current evidence does not support promoting cryptocurrencies as a low-risk alternative. In multi-asset portfolios, adding cryptocurrencies has almost no substantial impact on the minimum variance portfolio—the optimizer assigns a very low weight to cryptocurrencies due to risk control considerations.

After enhancing VADER with finance-specific terminology, the proportion of neutral sentiment judgements decreased from 49.6% to 47.8%. Without tuning parameters to the OOS results, fixed sentiment tilts improved the minimum-variance equity fund, especially in the contrarian direction; however, this remains exploratory OOS evidence rather than proof of a stable predictive relationship. Two robustness checks were also conducted: Ledoit-Wolf covariance shrinkage and a transaction-cost sensitivity analysis from 0 to 100 basis points. Finally, 50 headlines were manually labelled; agreement between the enhanced model and the human labels increased from 66% to 70%.

1. Product and investor proposition

SignalYield does not promise consistently outperforming the market. Instead, it addresses the real pain point of inconsistent and opaque information presentation by building a standardized comparison platform. Users can use a unified standard to view the out-of-sample performance of various funds, understand the latest asset allocation composition, freely combine different funds to build their own asset allocation plans, and refer to sector news and sentiment as supplementary background information. In terms of product interface design, we directly output pre-calculated result files instead of real-time calculation of raw data—this improves access speed, reduces deployment costs, and makes all results traceable and easy to audit. In terms of form, it resembles a systematic fund display platform, which could realize commercial value through a management fee model in the future; however, at this stage, it is only a prototype for educational purposes and does not constitute any investment advice.

In product design, we deliberately maintain the principle of "history is history, judgment is user-defined": performance tables truthfully present historical backtesting results; single-page introductions only objectively explain the characteristics of funds; allocation tools only linearly weight and combine existing returns; and sentiment results are clearly marked as "information for reference only." This is done to avoid presenting the interface in a too "definite" way, which could mislead users into thinking that the outcome is highly predictable.

2. Funds and backtest design

The product offers 12 systematically managed funds formed from three investable universes (equity-only, crypto-only and combined equity-crypto) and four construction methods. Applying the same methods to each universe separates the effect of asset selection from the effect of portfolio optimisation.

Asset universes. The equity-only funds invest in 50 shares across 10 sectors, the crypto-only funds invest in 10 cryptocurrencies, and the combined funds use all 60 assets. Crypto returns are first computed on crypto's seven-day calendar; the combined funds then use the equity trading calendar, so weekend-only crypto observations are not treated as separately tradeable fund returns.

Investment constraints. Every fund is long-only and fully invested: each weight is non-negative and all weights sum to one. Leverage, short selling and asset-level or crypto-exposure caps are not imposed in the baseline. This allows concentration to remain an observable economic outcome of the optimiser rather than being hidden by an arbitrary cap.

Portfolio construction. Except for equal weighting, the methods use the mean return vector μ and sample covariance matrix Σ estimated only from the expanding history available at each decision date. A small scale-proportional ridge is added to Σ for numerical stability, without changing the economic objective (Markowitz, 1952).

Equal weighting sets $w_i = 1/N$. It is the transparent, estimation-free benchmark against which the optimised funds are judged.

Minimum variance chooses the long-only weights that minimise $w' \Sigma w$. It targets portfolio risk without estimating expected returns.

Maximum Sharpe chooses weights that maximise $w'(\mu - r_f)/\sqrt{w' \Sigma w}$, with the daily risk-free rate fixed at zero. It can reward assets with high estimated returns, but is correspondingly sensitive to estimation error (Sharpe, 1966).

Risk parity chooses weights so that each asset contributes approximately $1/N$ of total portfolio variance, where $RC_i = w_i(\Sigma w)_i / (w' \Sigma w)$. This diversifies risk contribution rather than allocating equal dollars to every asset (Maillard, Roncalli and Teiletche, 2010).

Retaining equal weighting beside the three optimised methods is deliberate. With a short and noisy estimation history, errors in expected returns and covariances can outweigh the theoretical benefit of optimisation; a simple $1/N$ portfolio therefore provides a demanding and economically meaningful benchmark (DeMiguel, Garlappi and Uppal, 2009).

Walk-forward out-of-sample design. Each fund must accumulate approximately one year of common return observations before launch. After that point, the estimation window expands through time, retaining all earlier observations rather than repeatedly fitting a fixed rolling window.

Equity-only and combined funds require 252 equity-trading-day observations; crypto-only funds require 365 calendar-day observations. The same distinction is used for annualisation: 252 periods per year for equity and combined funds, and 365 for crypto-only funds.

At the first trading date of each month, parameters and target weights are estimated using data available through that date. The target becomes effective on the next available trading day, so every realised OOS return is strictly later than the observations used to form its applicable weight.

The first live dates are 5 January 2021 for the equity-only and combined funds and 2 January 2021 for the crypto-only funds. Each fund has 36 monthly decisions through 2023, giving 432 method-level rebalance records across the 12 funds; in every record, `estimation_end` is earlier than `effective_date`.

Once a target becomes live, holdings are allowed to drift with each asset's realised return until the next rebalance. Turnover is measured from the drifted pre-trade portfolio to the new target as 0.5 times the sum of absolute weight changes. Initial funding is assigned zero turnover because it is not a trade from an existing portfolio.

Fact-sheet performance is calculated only from live OOS returns. Annualised return is geometric, volatility is the sample daily standard deviation times the square root of the relevant annualisation factor, Sharpe uses a zero risk-free rate, and maximum drawdown is the largest peak-to-trough loss in growth of one dollar.

Baseline transaction costs are set to zero so the 12 funds isolate portfolio-construction effects. Turnover is still reported, and the separate robustness analysis applies fixed one-way costs without re-estimating the portfolios. Table 1 summarises the assumptions frozen before the OOS results were examined.

Design choice	Implementation	Reason
Initial / expanding window	252 observations for equity/combined; 365 for crypto, then expanding	Provides one year of history before launch and retains all earlier data

Rebalance	Monthly, estimated on the first trading date	Uses a consistent decision schedule across the 2021-2023 OOS period
No-look-ahead timing	Target weights become live on the next available trading day	Applicable returns are strictly later than the estimation data
Portfolio constraints	Long-only, fully invested, with no exposure caps	Creates an investable baseline while revealing optimiser concentration
Performance conventions	$r_f = 0$; 252/365 annualisation; baseline transaction costs = 0	States the gross-return comparison before separate cost sensitivity

Table 1. Frozen baseline backtest design, specified before OOS evaluation.

3. Out-of-sample fund results

Table A1 and Figures A1 and A4 provide the full 12-fund comparison. Cryptocurrency funds have the most volatile growth paths, whereas minimum-variance equity and combined funds are visibly more stable. Equal-weighted equities lead the equity methods, with an annualised Sharpe ratio of 0.86. This is economically plausible: with a short, changing sample, errors in estimated means and covariances can offset the theoretical benefits of optimisation. Equity maximum Sharpe performs worst among the equity methods and has the highest turnover, consistent with its sensitivity to estimated expected returns.

Figure A2 shows the drawdown contrast. Crypto risk parity and equal weight deliver annualised returns of 38.5% and 37.1%, respectively, but annualised volatility is approximately 80%-82% and maximum drawdowns exceed 80%. Even the crypto minimum-variance fund falls 75.6% from peak to trough, versus -17.9% for equity minimum variance. The combined and equity minimum-variance growth paths nearly overlap because the optimiser assigns almost no weight to crypto when minimising risk.

Figure A3 confirms the allocation mechanism across the 36 monthly rebalances. Crypto averages 16.7% in combined equal weight, 28.3% in combined maximum Sharpe, about 9.0% in combined risk parity and effectively 0.0% in combined minimum variance. Combined risk parity performs best among the combined methods, with a Sharpe ratio of 0.80; combined maximum Sharpe is weakest, at 0.44.

Illustrative fund	Return	Vol	Sharpe	Max DD
Combined Min Variance	7.0%	12.8%	0.59	-17.9%
Combined Risk Parity	13.2%	17.4%	0.80	-22.2%
Crypto Risk Parity	38.5%	80.1%	0.81	-80.6%
Equity Equal Weight	13.3%	16.1%	0.86	-20.2%

Table 2. Selected OOS results, 2021-2023; annualised, risk-free rate 0.

Table 2 condenses the selected OOS results and reinforces that fund selection should not focus on terminal return alone. Equal-weighted equities and combined risk parity offer a more favourable balance between return and drawdown, while any material crypto allocation requires acceptance of potentially severe losses. These are historical backtests, not forecasts, and three OOS years are insufficient to establish a permanent ranking.

4. News sentiment and structured-unstructured fusion

Sentiment processing retains original capitalisation and punctuation because VADER uses capital letters, exclamation marks and other emphasis markers when estimating intensity (Hutto and Gilbert, 2014). Headlines are scored individually and aggregated to a sector sentiment index by trading day, ticker and sector; a missing-news day is kept distinct from a neutral-sentiment day. To prevent same-day look-ahead, the index is lagged by one equity trading day: weekend news maps to Monday and can first influence Tuesday's position.

To make the model more familiar with financial contexts, we compiled 24 commonly used financial words/phrases, checked their sentiment tendencies and applicable scenarios one by one, and adopted them all, forming an expanded vocabulary table for the financial field. After adding this, the proportion of headlines judged as "neutral" decreased by approximately 1.76 percentage points out of 146,830 headlines—indicating that the model's sentiment recognition is more discriminative, rather than directly proving "more accurate predictions." The rationale behind this approach is that sentiment words in ordinary text can have entirely different meanings in a financial context; therefore, manual review and verification are essential (Loughran and McDonald, 2011).

Table 3 reports the completed manual validation of 50 sampled headlines (9 negative, 26 neutral and 15 positive), with average reviewer confidence of 3.78/5. Agreement between the enhanced model and human labels rises from 66% to 70%, and positive-class agreement improves from 66.7% to 80.0%; negative-class agreement remains only 33.3%. The exercise therefore supports a modest improvement, not a claim that the classifier is reliable for autonomous trading.

Model	Overall	Negative	Neutral	Positive
Vader Plain	66.0%	33.3%	76.9%	66.7%
Vader Enhanced	70.0%	33.3%	76.9%	80.0%

Table 3. Agreement with the completed 50-headline human review.

Figure A5 presents the sector sentiment index and its 20-day mean. Utilities have the highest average sentiment (0.197) and Financials the lowest (0.104). Technology averages about 26.5 headlines per day, compared with 5.3 for Real Estate, so coverage is uneven. The smoothed series reduces the risk that users over-interpret individual daily observations; the index remains contextual information, not a market forecast.

Table 4 and Figure A6 compare the equity minimum-variance base with fixed sector-level sentiment tilts based on the preceding 30-day sector sentiment Z-score. The three pre-specified variants are base, momentum and contrarian; no direction or parameter is selected by searching the OOS results. Sharpe rises from 0.59 for the base to 0.63 for momentum and 0.74 for contrarian. Contrarian also raises annualised return from 7.0% to 9.9% and improves maximum drawdown from -17.9% to -16.6%. This is an exploratory association, not causal

evidence: news may reflect information already incorporated into prices and may be affected by media-selection bias (Tetlock, 2007).

Strategy	Ann. return	Ann. vol.	Sharpe	Max drawdown
Base minimum variance	6.96%	12.75%	0.591	-17.93%
Momentum tilt ($\lambda=+1$)	7.94%	13.60%	0.630	-19.81%
Contrarian tilt ($\lambda=-1$)	9.88%	14.06%	0.741	-16.65%

Table 4. Equity minimum-variance base versus fixed sentiment tilts, OOS 2021-2023.

5. Innovation and robustness

5.1 Ledoit-Wolf covariance shrinkage

The minimum variance and maximum Sharpe strategies are highly dependent on the covariance matrix—therefore, we introduce the Ledoit-Wolf covariance shrinkage estimator, which automatically adjusts the shrinkage intensity within each expansion window, without using out-of-sample data for parameter tuning throughout the process (Ledoit and Wolf, 2004). The original baseline results remain unchanged, serving only as a parallel control.

Family	Method	LW Sharpe	Change
Combined	Max Sharpe	0.42	-0.013
Combined	Min Variance	0.58	-0.014
Crypto	Max Sharpe	0.62	+0.038
Crypto	Min Variance	0.77	+0.103
Equity	Max Sharpe	0.63	+0.015
Equity	Min Variance	0.58	-0.006

Table 5. Ledoit-Wolf versus sample-covariance OOS Sharpe, 2021-2023.

Table 5 and Figure A7 show that covariance shrinkage is not a universal improvement. It has the largest benefit in the high-variance crypto universe, where the minimum-variance Sharpe ratio rises by 0.103, while its effects on equities and the low-volatility combined portfolios are mixed and small. The result is therefore conditional: robust covariance estimation is more valuable where sampling noise is severe, but it does not guarantee better OOS performance.

5.2 Transaction-cost sensitivity

Figure A8 applies one-way transaction costs of 0, 10, 25, 50 and 100 basis points without changing the target weights. Turnover is computed from the drifted pre-trade portfolio to the next target, so even a fixed-target equal-weight fund incurs rebalancing costs after relative prices move. Costs have the greatest impact on the highest-turnover strategy, equity maximum Sharpe, whose cumulative turnover is 4.79 times capital; at 50 basis points its Sharpe ratio falls by 0.034.

Figure A8 also illustrates why a theoretically attractive strategy can disappoint after implementation when estimates are unstable and trading is frequent. The sensitivity curve is deliberately simplified: a uniform one-way rate does not capture bid-ask spreads, market impact, taxes or asset-specific liquidity, so it is a robustness diagnostic rather than an executable cost forecast.

6. App and investor journey

The product interaction page is divided into six pages: Home Introduction, Fund Comparison, Fund Details Page, Portfolio Allocation, Sentiment Reference, and Robustness Check. The homepage first explains the scope of application and risk warnings; the comparison page allows filtering of all 12 funds; the details page centrally displays returns, drawdowns, and the latest portfolio composition; the portfolio allocation page allows users to freely combine funds and view the combined performance in real time (weighted at 100%, calculated only within a common time interval); the sentiment page displays the sector index and strategy comparison results; and the robustness check page publicly presents how covariance is calculated and costs are deducted, without hiding them in the appendix.

The technical architecture is deliberately lightweight: the application interface reads pre-computed result files rather than downloading raw data, running NLTK or re-optimising portfolios at request time. This improves speed and makes every displayed result traceable to the reproducible pipeline. The code is hosted in the public GitHub repository https://github.com/RUOYUNWU-ui/z5652591_projectB on the main branch. The final hand-in must also include and verify the live Streamlit app URL.

7. Critical reflection and recommendations

First, always maintain a simple benchmark. Equal weighting performs best in equities; risk parity shows robust performance and a turnover rate far lower than the maximum Sharpe ratio. If a formal version is released, users should be explicitly informed that "optimization methods involve estimation risks," rather than implying that "algorithm optimization is necessarily better."

Second, cryptocurrency risks must be clearly disclosed separately. Even with "volatility reduction" optimizations, drawdowns generally exceed 75%. The product should set cryptocurrency configuration as a "separate risk module," requiring users to be clearly aware of the potential for substantial losses before selection. Contextualized risk warnings are recommended.

Third, sentiment models are currently only suitable for reference. Overall improvement is limited, and negative identification remains inaccurate—to truly apply them to strategies, a larger-scale, independently labeled validation sample is needed, with sector-specific problem identification investigated one by one.

Three follow-up recommendations follow. First, extend the sample before tuning parameters: three OOS years may represent only one market regime, and optimisation against this period would invite overfitting. Second, define a consistent fee methodology before adding differentiated cost models; execution complexity should be introduced only when it can be validated. Third, reserve a genuinely untouched validation period for the sentiment strategy. Repeatedly changing parameters or signal directions merely to improve the OOS Sharpe ratio is specifically not recommended because it disguises backtest overfitting rather than resolving it (Bailey et al., 2014).

Appendix A. Required and extension exhibits

**Table - OOS performance metrics across 12 systematic funds
2021-01 to 2023-12; rf=0, baseline transaction cost=0**

Fund	Ann. return	Ann. vol	Sharpe	Max drawdown	Turnover
Combined - Equal Weight	15.5%	21.5%	0.78	-27.7%	3.9%
Combined - Maximum Sharpe	9.7%	35.3%	0.44	-52.6%	15.6%
Combined - Minimum Variance	7.0%	12.8%	0.59	-17.9%	3.1%
Combined - Risk Parity	13.2%	17.4%	0.80	-22.2%	3.3%
Crypto-only - Equal Weight	37.1%	81.7%	0.80	-81.7%	5.8%
Crypto-only - Maximum Sharpe	15.1%	79.4%	0.58	-84.9%	6.9%
Crypto-only - Minimum Variance	25.0%	65.1%	0.67	-75.6%	2.3%
Crypto-only - Risk Parity	38.5%	80.1%	0.81	-80.6%	5.8%
Equity-only - Equal Weight	13.3%	16.1%	0.86	-20.2%	2.8%
Equity-only - Maximum Sharpe	12.5%	23.6%	0.62	-37.9%	13.7%
Equity-only - Minimum Variance	7.0%	12.8%	0.59	-17.9%	3.1%
Equity-only - Risk Parity	10.6%	14.9%	0.75	-19.4%	2.6%

Table A1. OOS performance metrics for all 12 funds, 2021-2023; annualised, rf=0.

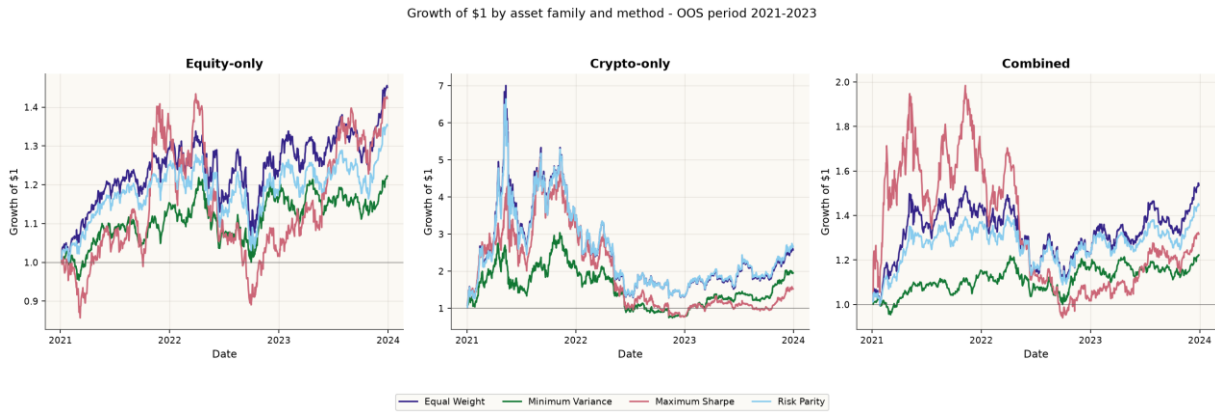


Figure A1. Growth of \$1 across funds and methods, OOS 2021-2023.

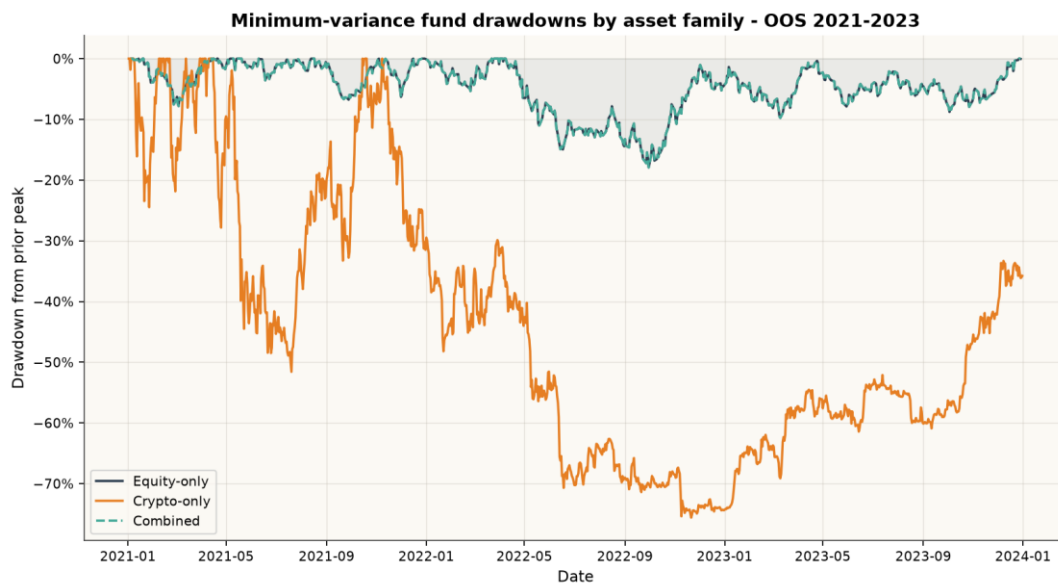


Figure A2. Minimum-variance drawdown by asset family, OOS 2021-2023.

Combined-fund sector and crypto weights over time - 36 monthly OOS rebalances



Figure A3. Combined-fund sector and crypto weights across 36 monthly rebalances.

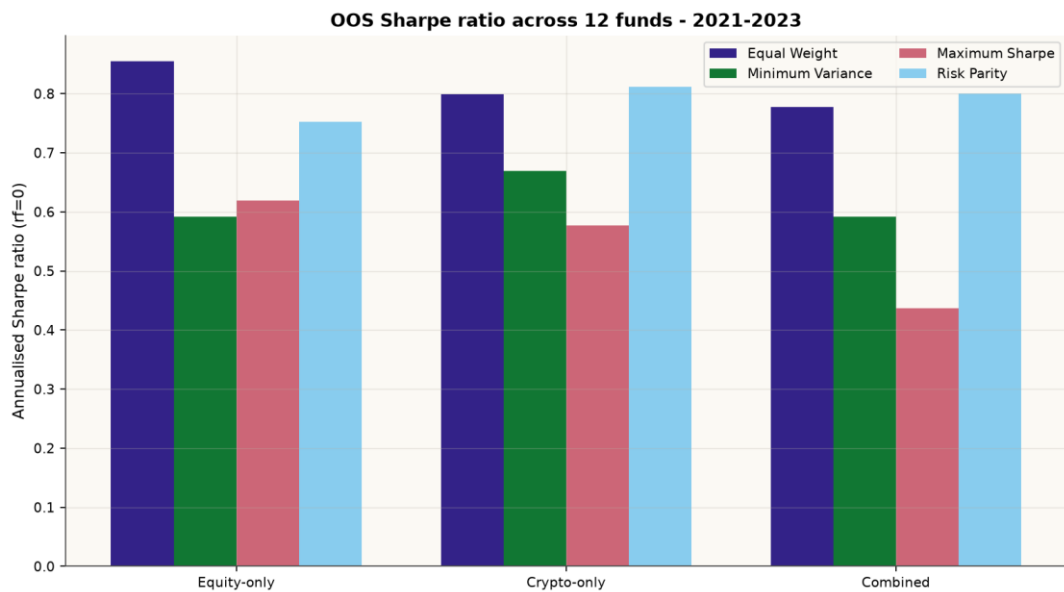


Figure A4. Annualised Sharpe ratios across the 12 funds, OOS 2021-2023.

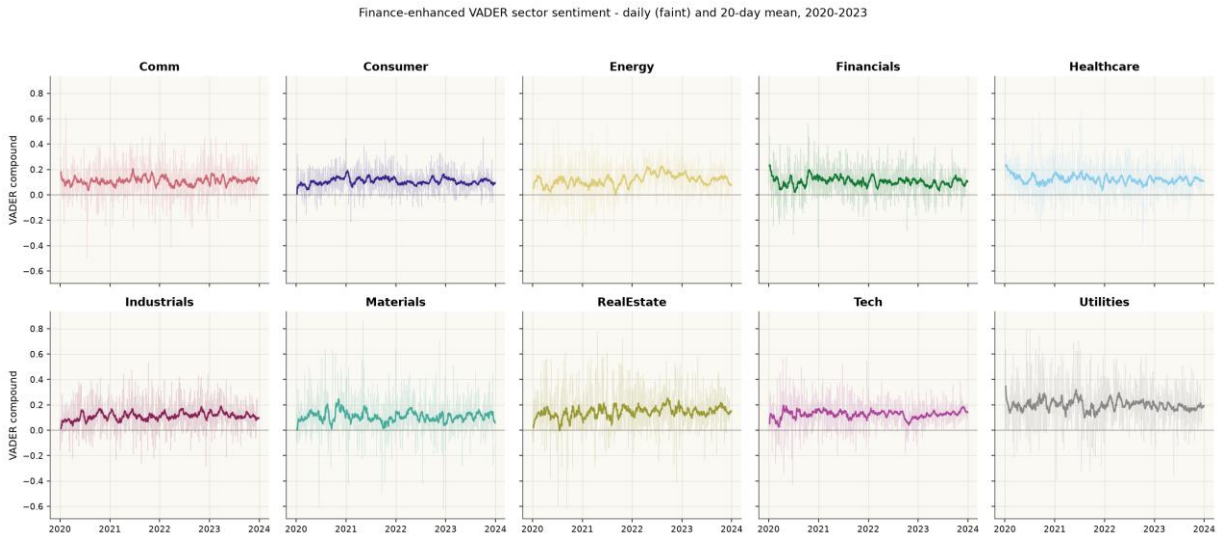


Figure A5. Finance-enhanced VADER sector sentiment, daily and 20-day mean, 2020-2023.

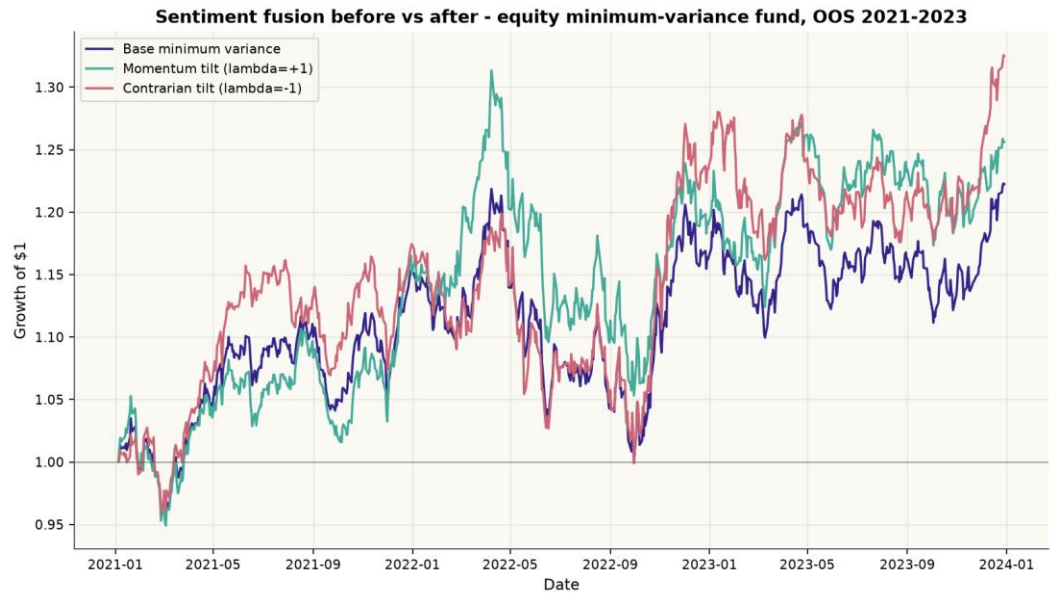


Figure A6. Equity minimum-variance base versus fixed sentiment tilts, OOS 2021-2023.

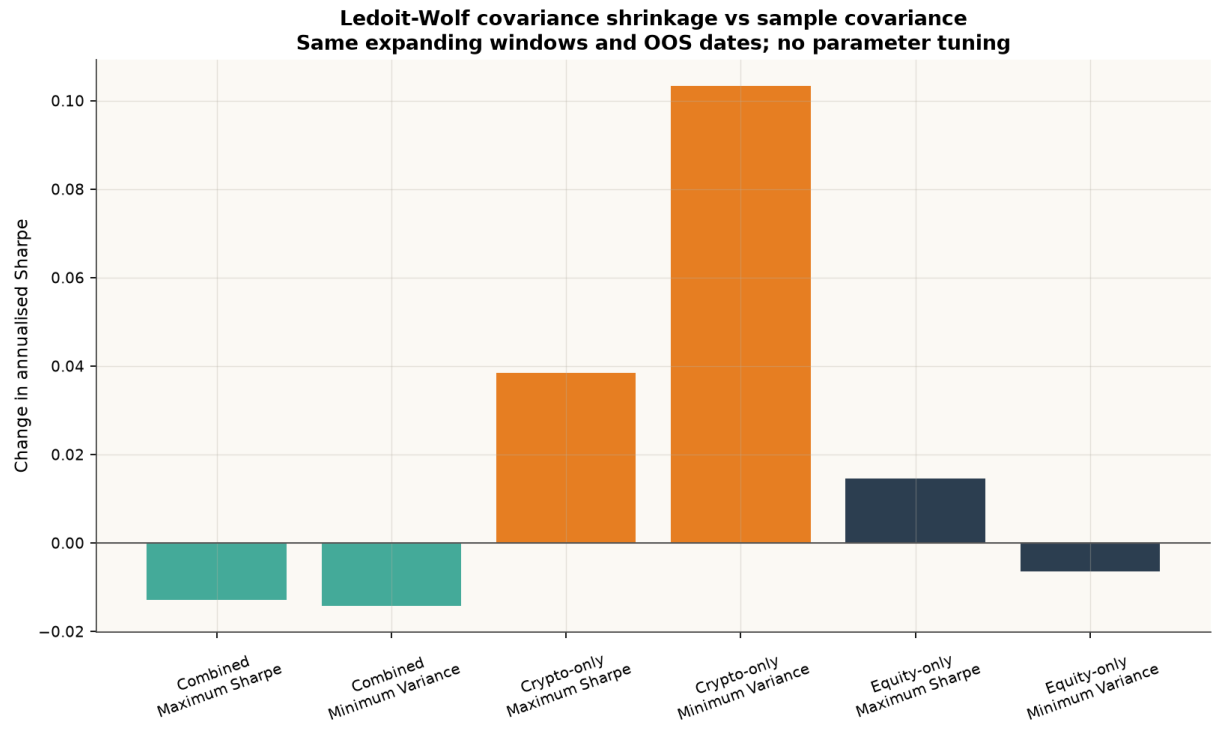


Figure A7. OOS Sharpe change under untuned Ledoit-Wolf covariance shrinkage.

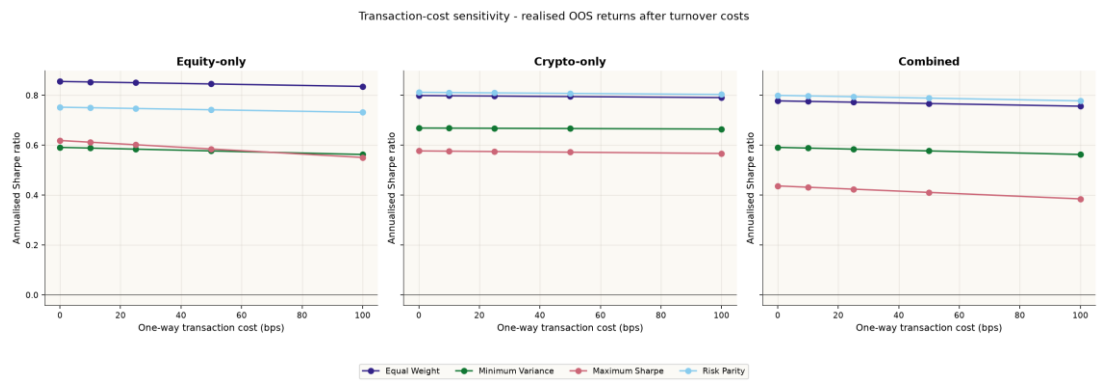


Figure A8. OOS Sharpe sensitivity to one-way costs from 0 to 100 bps.

Appendix B. Reproducibility and AI transparency

Table B1 lists the key evidence paths. The complete evidence is rebuilt by running python scripts/run_part_b.py from the repository root; it writes the app data, report tables and figures. Automated tests cover portfolio timing, sentiment lagging, fusion invariants, robustness and all six app pages. scripts/check_handin.py audits required files and frozen data access. Detailed prompts, corrections and AI limitations are recorded in ai/prompt_log_part_b.md and report/QA_REPORT.md.

Evidence	Location
12-fund returns and weights	results/data/fund_returns.csv; fund_weights.csv
Sector sentiment	results/data/sector_sentiment_index.csv
Fund metrics	results/tables/performance_metrics.csv
Robustness tables	results/tables/shrinkage_comparison.csv; transaction_cost_sensitivity.csv
Completed manual review	sentiment_manual_review_template.csv; sentiment_manual_review_validation.csv

Table B1. Key reproducible evidence paths.

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